are now giving way to novel media and incubation conditions. One approach is to use unusual electron donors and acceptors. For instance, a novel isolate from Mono Lake (California) sediments was cultivated only when anoxic conditions with arsenite as the electron donor and nitrate as the electron acceptor were provided. It has also become clear that growing novel microbes requires patience. Microbiologists are most familiar with doubling times in the order of hours or, at the most, a few days, when in fact, this is not always the case as noted in **Microbial Diversity & Ecology 26.1**. Furthermore, microbiologists have been trained to equate turbidity in liquid culture with growth. This is not always true, as some bacteria stop dividing (i.e., reach stationary phase) at cell densities too low to appear turbid. Lastly, in recognition that the growth of one microbe might depend on the product(s) of another, techniques have been developed that allow bacteria to exchange secreted metabolites and signaling molecules through filters or gels.

A variety of new culture techniques have been applied that address specific needs of microorganisms growing in diverse environments. Efficiency can be greatly improved by using high-throughput methods. One popular approach is the **extinction culture technique,** so named because natural samples are first microscopically examined and then diluted to a density of 1 to 10 cells—that is, diluted to extinction, much like in the MPN assay. Multiple 1-milliliter cultures are established in 48-well

MICROBIAL DIVERSITY & ECOLOGY

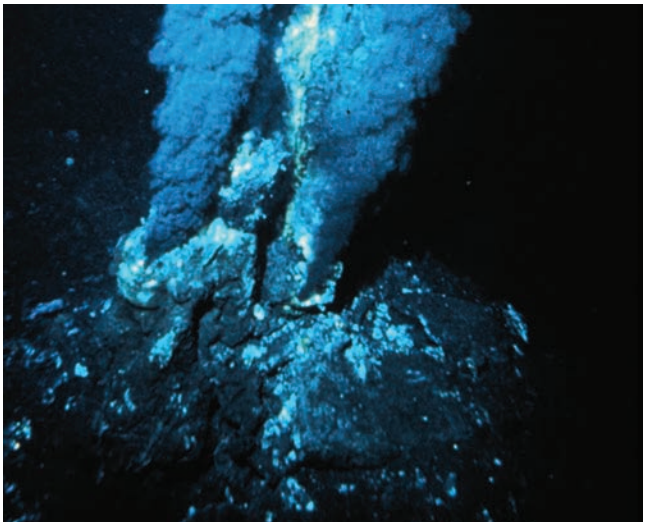
26.1 Patience, Hard Work, Luck, and the Evolution of Eukaryotes

One of the most vexing questions in evolution concerns the origin of eukaryotes. One hypothesis posits that eukaryotes evolved from ancestors of so-called Asgard archaea. It is suggested that these ancient archaeal ancestors and their endosymbiotic bacteria gave rise to the earliest eukaryote with mitochondria of bacterial origin (*see section 1.2*). But piecing together evolutionary events is a little like developing a case in a court of law. There are two types of evidence. The first is circumstantial evidence such as a person who resembles the suspect was seen carrying a weapon. The microbial equivalent might be the genome of a living archaeon that encodes proteins with functions similar to eukaryotes (e.g., genes involved in replication and transcription). The second is direct evidence: the suspect was caught engaging in illegal activity. Here the microbial parallel would be a genome of a living archaeon that encodes so-called signature eukaryotic genes— genes that were thought to be uniquely eukaryotic, such as those involved in membrane trafficking vesicle formation, ubiquitin, and cytoskeleton formation.

In 2010, tantalizing circumstantial evidence was found. Metagenomic analysis of samples taken from a deep-sea hydrothermal vent (**box figure**) called Loki's castle revealed most of the genome of a novel archaeon dubbed Lokiarchaeum. What was so intriguing about this metagenome was that in addition to archaeal genes, it encoded signature eukaryotic proteins as well. However, this evidence was considered inadmissible in the court of evolutionary science because the risk of contamination by eukaryotic genes was considered too high. After all, these genes had never been seen in bacteria or archaea before.

As the evolutionary implications of the Lokiarcheaum metagenome was being publicly debated, Hiroyuki Imachi's laboratory in Japan had placed some deep-sea hydrothermal vent sediment in a bioreactor that mimicked the vent environment (i.e., a sophisticated enrichment culture). Then they waited. They waited until 5 years had passed to take a sample from the bioreactor, which they incubated for yet another year in separate enrichment cultures. Finally, in 2019 they had a co-culture of Lokiarchaeum and a methanogenic archaeon. They named the former *Candidatus* Prometheoarchaeum syntrophicum strain MK-D1. Turns out *Ca*. P. syntrophicum has a doubling time of 2–3 weeks.

Most importantly, Dr. Imachi's group sequenced the genome of *Ca*. P. syntrophicum, thereby providing direct evidence that it does indeed encode signature eukaryotic genes. By providing the first lokiarchaeal genome, *Ca*. P. syntrophicum is accepted as the closest, cultured archaeal relative of eukaryotes. And although there is about 2 billion years between *Ca*. P. syntrophicum and the archaeon that gave rise to eukaryotes, we are closer than ever in being able to create and test hypotheses about the origin of eukaryotes. ◀ *Asgardarchaeota are close relatives of eukaryotes (section 19.2)*



OP. Rona/AR/National Undersea Research Program (NURP)/NOAA

microtiter dishes. After incubating for the desired time under appropriate conditions, the samples are stained and examined microscopically to screen for growth. In many cases, cultures that do not appear turbid will nonetheless show signs of viable microbial growth and can be transferred to fresh medium. It is important that the initial culture has fewer than 10 cells; additional transfers and dilutions can ensure that a culture arising from a single cell (i.e., clonal growth) eventually is obtained.

Recent progress in cataloging microbes that make up the human microbiome has impelled the development of the high-throughput cultivation technique called culturomics. **Culturomics** involves at least three steps: First, samples of a mixed assemblage of microbes are used as inoculum for hundreds of different culture media incubated under a variety of physical conditions (e.g., temperature, pH, and oxygen concentration). Microorganisms from colonies that grow are then identified by mass spectrometry or small subunit (SSU; 16S for bacteria and archaea, 18S for eukaryotic microorganisms) rRNA gene sequencing. From these initial cultures, larger cultures can be started—that is, cultures are scaled up and in some cases, organisms are grown in pure culture. Even if pure culture is not possible, these microbes can be the source for genome sequencing by single-cell genomics. *Single-cell genomic sequencing (section 18.2); Proteomics explores total cellular protein (section 18.5)*

Flow cytometry has become a commonly used technique to isolate single cells from a mixed population. This can provide isolated cells to start cultures. In **flow cytometry,** cells are tagged with a fluorescent dye and injected into a flowing stream of fluid. As shown in **figure 26.3**, as the diameter of the stream narrows, one cell at a time is forced through a thin tube. Each cell is detected by a laser beam. The laser excites the fluorescent label and light emitted by each cell is collected by a detector that converts the signal made by the cell into a voltage pulse proportional to the size of the cell. Some of the cell's emitted light scatters to the side; this is a function of morphological complexity of the cell. This is also detected and quantified. Thus based on cell size and morphology, flow cytometry can detect different populations of cells. This enables an additional function, **fluorescence activated cell sorting (FACS).** Here, as the stream leaves the detector, vibration causes the fluid to break into single droplets with one cell per drop. Cells become electrostatically charged as they pass the detector. The charge is used to guide each cell into a collection vessel. Thousands of cells per second pass through the laser beam, making "flow" an extremely efficient means to count, sort, and collect cells.

Comprehension Check

1. When might a microbe be considered VBNC?
2. How would you determine the ratio of dead to living microbial cells in a soil sample?
3. Describe the rationale upon which the extinction culture technique is based.
4. How might you use FACS to obtain bacterial and archaeal cells for analysis from lake water?

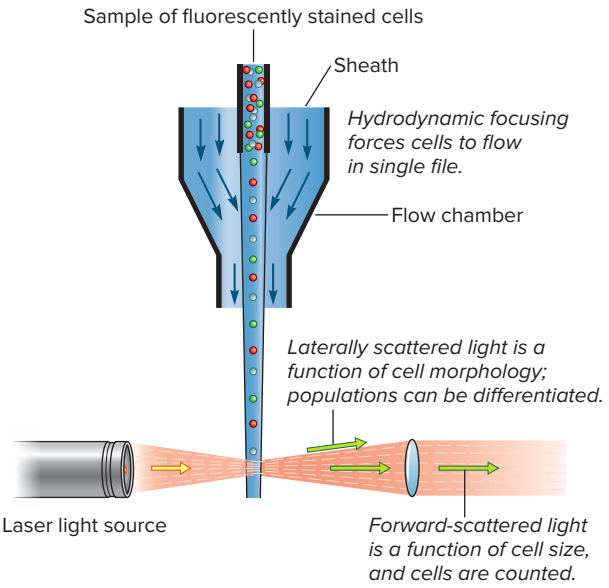


Figure 26.3 Flow Cytometry.

MICRO INQUIRY *What happens to the flow stream in fluorescence activated cell sorting (FACS)?*

26.2 Microbial Identification Is Largely Based on Molecular Characterization

After reading this section, you should be able to:

- **a.** Construct a concept map describing the approaches commonly used to determine taxonomic classification
- **b.** Compare and contrast nucleotide sequencing and nonsequencing-based molecular approaches used in microbial identification
- **c.** Select an appropriate technique to identify a microbial genus, species, and strain
- **d.** Predict the basic biological as well as public health implications of microbial taxonomic identification

Once a microbe has been isolated from its natural environment, an essential question is, "What is its taxonomic classification?" Microbiologists use a variety of approaches to identify a microbe's genus and species. Many, but not all, require that the organism be grown in pure culture. For clarity, we divide these approaches into two groups: classical and molecular. The most durable identifications are those based on a combination of approaches.

Classical Characteristics

Classical approaches to taxonomy make use of morphological, physiological, and ecological characteristics. These characteristics have been employed in microbial taxonomy for many years and form the basis for phenetic (phenotypic) classification. When

used in combination, they are quite useful in routine identification of well-characterized microbes.

Morphological features such as cell shape, size, and staining behavior (for example, Gram positive or negative) are important in microbial taxonomy for many reasons. Morphology is easy to study and analyze, particularly in eukaryotic microorganisms. In addition, morphological comparisons are valuable because structural features depend on the expression of many genes and are usually genetically stable. Thus morphological similarity can be a good indication of phylogenetic relatedness.

Physiological and metabolic characteristics are useful because they are directly related to the nature and activity of microbial proteins. For instance, the detection of specific end products of fermentation in a newly discovered microorganism reveals the presence of specific catabolic enzymes and the genes that encode them. Therefore analysis of characteristics, such as energy metabolism and nutrient transport, provides an indirect comparison of microbial genomes.

The ability of a microorganism to colonize a specific environment is of taxonomic value. Some microbes may be similar in many other respects but inhabit different ecological niches, suggesting they may not be as closely related as first suspected. Some examples of taxonomically important ecological properties are life cycle patterns; the nature of symbiotic relationships; the ability to cause disease in a particular host; and habitat preferences such as requirements for temperature, pH, oxygen, and osmotic concentration. ◀ *Environmental factors affect microbial growth (section 7.5);* ▶ *Many types of microbial interactions exist (section 27.1)*

Molecular Characteristics

It is hard to overstate how the study of DNA, RNA, and proteins has advanced our understanding of microbial evolution and taxonomy. We first describe two biochemical approaches to microbial identification followed by genetic approaches.

Biochemical Characteristics

Among the more useful biochemical characteristics used in microbial taxonomy are bacterial fatty acids, which can be analyzed using a technique called *f*atty *a*cid *m*ethyl *e*ster (FAME) analysis. A fatty acid profile reveals differences in chain length, degree of saturation, branched chains, and hydroxyl groups. Microbes of the same species will have identical fatty acid profiles, provided they are grown under the same conditions; this limits FAME analysis to only those microbes that can be grown in pure culture. Also, because the identification of a species is done by comparing the results of the unknown microbe in question with the FAME profile of other, known microbes, identification is only possible if the species in question has been previously analyzed. Nonetheless, FAME analysis is particularly important in public health, food, and water microbiology. ▶ *Identification of microorganisms from specimens (section 36.2); Microbiology of food (chapter 40); Purification and sanitary analysis ensure safe drinking water (section 42.1)*

Advances in mass spectrometry (MS) have resulted in the fast and accurate identification of bacteria based on the presence of specific, highly abundant proteins. The specific type of MS used is called *m*atrix-*a*ssisted *l*aser *d*esorption/*i*onization-*t*ime *o*f *f*light (MALDI-ToF). MALDI-ToF enables the analysis of complex biomolecules that could not previously be studied by MS. The material to be analyzed is dried on a sample holder (called a target) and then mixed with a molecular film called a matrix. When a UV laser beam strikes the sample target, the matrix helps stimulate release of the sample from the surface; this is *matrix-assisted desorption*. Once the matrix-sample is released (desorbed), the matrix transfers protons to the sample, which then becomes ionized; this is *matrix-assisted ionization*. The ionized, biomolecules are then "flown" across a space within the instrument; the time taken for a molecule to fly from one side to the other (*time of flight*) is used to determine the mass of each molecule.

In its simplest form, MALDI-ToF identification of bacteria involves the transfer of whole cells from a single colony grown under specific conditions to a sample target. Once dried, a matrix is deposited and the bacterial samples are analyzed. MALDI-ToF yields the masses of highly abundant bacterial proteins. Like FAME analysis, each experimentally derived protein profile must be matched to a protein profile from a known bacterium. MALDI-ToF is becoming increasingly important in medical microbiology laboratories where the same strains of organisms are regularly encountered. Like FAME, microbes must be grown under specific conditions and MALDI-ToF cannot be used to identify newly discovered microbes. ◀ *Proteomics explores total cellular protein (section 18.5)*

Signature Sequences

As introduced in chapter 1, **small subunit ribosomal RNAs (SSU rRNAs)** are commonly used to infer microbial phylogenies and make taxonomic assignments at the genus level. These molecules possess **oligonucleotide signature sequences,** short, conserved nucleotide sequences specific for phylogenetically defined groups of organisms *(see figure 1.8).*

The ability to amplify regions of rRNA genes (rDNA) by the polymerase chain reaction (PCR) and sequence the DNA using next generation sequencing (NGS) techniques has greatly increased the efficiency by which SSU rRNA sequences are obtained. PCR amplifies rDNA from the genomes of different organisms because conserved nucleotide sequences that flank the target nucleotides can be used to reveal the microbe's identity. In practice, this means that these PCR primers amplify rDNA from both cultured and uncultured microbes. Any uncultivated microorganism identified solely on its nucleic acid sequence (or other observable, quantifiable phenotype) is termed a **phylotype.** ◀ *Polymerase chain reaction amplifies targeted DNA (section 17.2); DNA sequencing by synthesis (section 18.1)*

The use of SSU rRNA sequences as a taxonomic marker has been validated as a molecular measure of relatedness. When SSU rRNA sequence data is compared with traditional measures

of assigning species, organisms of the same species show at least 98.65% sequence identity in their SSU rDNA sequences. This means that when comparing the SSU rRNA sequences of two organisms, differences at more than 20 positions of a ~1,540 base SSU rRNA indicate that they probably derive from different species.

Signature sequences are present in genes other than those encoding rRNA. Many genes have nucleotide insertions or deletions of specific lengths and sequences at fixed positions. A particular nucleotide sequence that is inserted or deleted may be found exclusively among all members of a phylum. These taxon specific insertions and deletions are called conserved **indels** (for *in*sertion/*del*etion). Indels are particularly useful in phylogenetic studies when they are flanked by conserved regions. In such cases, changes in the signature sequence cannot be due to sequence misalignments. Alignment of the inferred amino acid sequences reveals indels useful for phylogenetic analysis.

Whole-Genome Comparison

As the field of taxonomy shifts to whole-genome sequencing (WGS) for strain identification and taxonomic classification, new quantitative measures of relatedness are being developed and evaluated. As with any metric, standardization of methods and interpretations are required. These processes are ongoing. An initial step to identify a newly discovered microbe is SSU rRNA comparison, which provides confident identification to the genus level. ◀ *Genome sequencing (section 18.2)*

Identification to the species level requires detailed gene-by-gene comparison of the new isolate and type strains of its closest relatives. **Average nucleotide identity (ANI)** is now widely considered to be the standard for species identification. This technique uses pairwise alignments between all sequences shared between two genomes and calculates the fraction of identical nucleotides. ANI values for two genomes of the same species should be at least 95 to 96%.

ANI is replacing a biochemical technique called **DNA-DNA hybridization (DDH).** DDH is performed by mixing genomic DNA extracted from two strains. The mixture is heated until denaturation occurs, and then slowly cooled to allow renaturation. Noncomplementary regions remain unpaired, and the degree of renaturation is calculated. The results of this assay depend on the quality of the extracted DNA and other factors. By contrast, bioinformatics software quickly calculates digital DDH values using WGS data, although this technique is less commonly used than ANI.

Another biochemical technique gone digital is determination of **G + C content.** This metric is a simple percentage of the bases in DNA that are G + C, and it is readily determined computationally from WGS data. Organisms range from around 30 to 80% G + C. Despite the wide range of variation, the G + C content of strains within a species is constant and varies little within a genus.

Subspecies and Strain Identification

For many applications, identification to levels below species is required. Typically genes that evolve more quickly than those that encode rRNA must be analyzed. The technique called **multilocus sequence typing (MLST)** compares the sequences of several conserved housekeeping genes (**figure 26.4**). At least five genes are examined to avoid misleading results that can arise through horizontal gene transfer. Because many different versions, or alleles, of each gene exist, the finding that two microbial isolates share the same alleles for multiple genes is strong evidence that the two isolates are closely related, perhaps even the same strain. MLST is now often performed from *w*hole *g*enome sequences, where it is termed wgMLST. The availability of wgMLST data enables extended gene-by-gene comparison.

Highly conserved and repetitive DNA sequences present in many copies in the genomes of most Gram-negative and some Gram-positive bacteria can also serve in microbial identification. Three families of repetitive sequences are typically used for microbial identification: the 154 bp BOX elements, the 124–127 bp *e*nterobacterial *r*epetitive *i*ntergenic *c*onsensus (ERIC) sequence, and 35–40 bp *r*epetitive *e*xtragenic *p*alindromic (REP) sequences. Oligonucleotide primers are used to amplify the repetitive sequences by PCR. Different primers are used for each type of repetitive element and the results are classified as arising from BOX-PCR, ERIC-PCR, or REP-PCR. In each case, the amplified fragments from many microbial samples can be separated and visualized by electrophoresis. Each lane of the gel corresponds to an individual bacterial isolate and the pattern created by many samples resembles a UPC bar code. Using pattern recognition software,

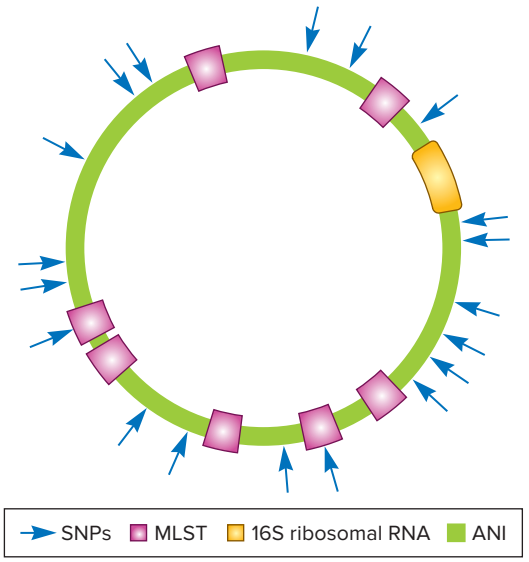


Figure 26.4 Genome Coverage of Genetic Taxonomic Approaches. A bacterial or archaeal genome contains at least one locus encoding 16S rRNA. MLST samples multiple genes throughout the genome. SNP analysis compares more, but shorter, nucleotide sequences that span the entire genome. ANI surveys the entire genome.

the bar code pattern enables phylogenetic analysis. This enables identification to the level of species, subspecies, and often strain, and is frequently used to identify animal (including human) and plant pathogens.

To survey a large fraction of the genome, **single nucleotide polymorphisms** (**SNPs,** pronounced "snips") are identified in specific genes, intergenic regions, or other noncoding regions. Originally developed to analyze human DNA, SNP analysis targets specific regions because they are normally conserved, so single base pair differences reveal evolutionary change. SNP analysis shares features with analysis of **restriction fragment length polymorphism (RFLP).** This technique identifies differences in restriction endonuclease digestion patterns, which reflect individual base pair changes because restriction enzymes recognize specific nucleotide sequences. When this analysis is applied to the gene encoding the SSU rRNA, it is termed **ribotyping.** With the advent of WGS, these techniques are increasingly performed in silico.

Comprehension Check

1. What are the advantages of using each major group of characteristics (morphological, physiological and metabolic, ecological, and molecular) in classification and identification? Give examples of each type of characteristic.
2. List the steps you would take to identify an organism just isolated in pure culture. What criteria would tell you if it is a new species?

26.3 Assessing Microbial Populations

After reading this section, you should be able to:

- **a.** Explain why microbial communities are often investigated in situ
- **b.** Describe why, when, and how FISH and CARD-FISH are used
- **c.** Describe the use of phylochips to assess microbial communities
- **d.** Discuss the role of metagenomics in assessing microbial diversity
- **e.** Predict which techniques would be appropriate for assessing the microbial community in soil or seawater

Microbial ecologists investigate natural communities in soils or waters, or in association with other organisms, including humans. Diversity is a measure of the number of genera, species, or ecotypes in an ecosystem. *Species richness* describes the number of different species in an environment; microbes can be evenly distributed, follow gradients, or inhabit patches within an ecosystem. Diversity, richness, and distribution reflect the various types of metabolic strategies used in a single habitat: What carbon sources and electron donors and acceptors are used by the different microbes? A microbial **population** is a group of microorganisms within a single species or other taxon in an ecosystem. When a group of microbes is defined by its physiological activity, it is called a **guild,** in the same way that a guild of people often represents their occupation, for example, a carpenters' guild. A microbial example would be a guild of nitrifying microbes in a soil sample. Finally, all microorganisms that share a common habitat comprise a **microbial community.**



Pixtal/age fotostock

Staining Techniques

The most direct way to assess microbial diversity is to observe microbes in nature. This can be carried out in situ using immersed microscope slides or electron microscope grids placed in a location of interest, which are then recovered later for observation. Samples taken from the environment often are examined in the laboratory using cellular stains or fluorescent dyes that stain specific molecules. Individual cells can then be counted and their concentration calculated. The fluorescent stains DAPI (4′,6-*dia*mido-2-*p*heny*l*indole) and acridine orange are commonly used to visualize microbes in environmental, food, and clinical samples (**figure 26.5**). These dyes specifically label nucleic acids (i.e., both DNA and RNA), and little sample preparation is required prior to their application. The nucleic acid dye

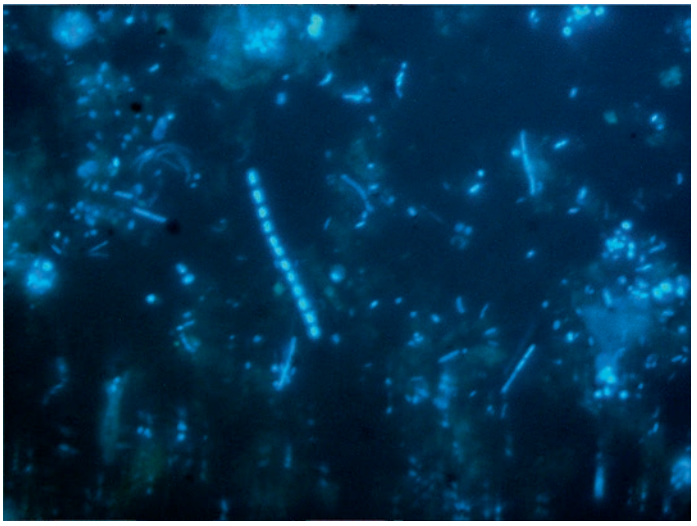


Figure 26.5 DAPI Stains Nucleic Acids. A microbial assemblage from the North Sea off the Belgian coast as visualized following DAPI staining.
David C. Gillan/Mons University, Belgium

SYBR Green is also used, in part because the signal is so bright virus particles can be seen despite their small size.

DNA Hybridization Techniques

Nucleic acid staining allows one to count the total number of microbes, but microbiologists often want to enumerate a specific genus or type of microbe. In these cases, a technique called ***f*luorescent *i*n *s*itu *h*ybridization (FISH)** can be used. Here fluorescently labeled, single-stranded oligonucleotides, called **probes,** known to be specific to the microbe of interest are used to label natural samples (**figure 26.6**). SSU rRNA signature sequences are commonly used probes. To perform FISH, microbial cell membranes are made permeable so the probe can be internalized. Once inside the cell, it hybridizes to complementary nucleotide sequences. A probe that does not bind is washed off, whereas the fluorescence of a bound probe reveals the organisms of interest. Probe fluorescence is visualized by **epifluorescence microscopy** (**figure 26.7**). Alternatively, individual cells can be separated and counted (but not visualized) by flow cytometry (figure 26.3). *Fluorescence microscopes use emitted light to create images (section 2.2)*

Depending on the probe used, FISH can identify a particular species or strain. This has made it a popular tool in clinical diagnostics and food microbiology, as well as microbial ecology. For example, FISH was used to discover the association between the archaea *Nanoarchaeum equitans* and *Ignicoccus hospitalis* (*see figure 19.6*).

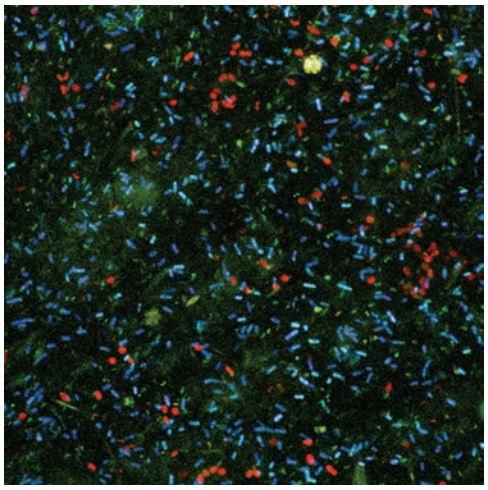


Figure 26.7 The Use of FISH to Study Microbial Populations. In this study, 16S rRNA probes tagged with different color fluorescent labels were used to study the microbial diversity within an earthworm gut.

Seana Davidson

MICRO INQUIRY *Under what circumstances would one use epifluorescence microscopy, rather than flow cytometry, and vice versa?*

Sometimes when FISH is used in natural samples, the fluorescent signal is not bright enough to be detected microscopically. This is because microbes living in these environments often grow quite slowly. Thus rather than having tens of thousands of

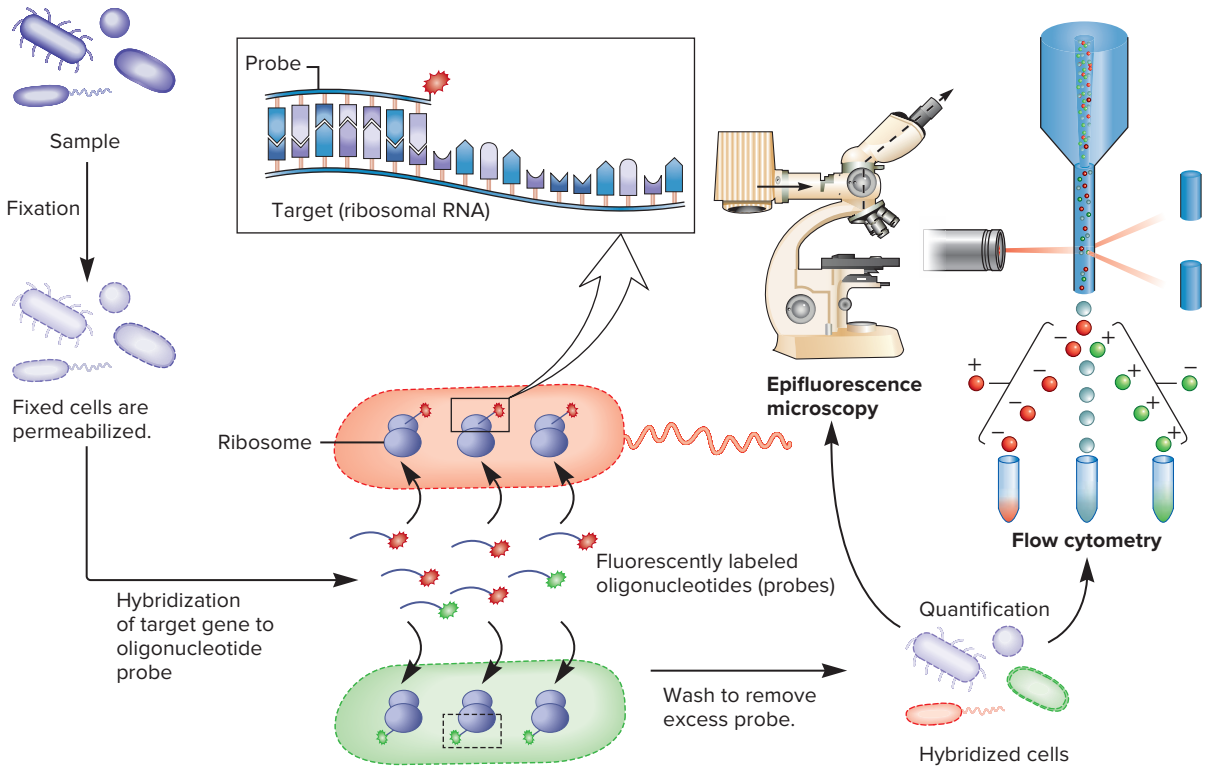


Figure 26.6 Fluorescence in Situ Hybridization (FISH).

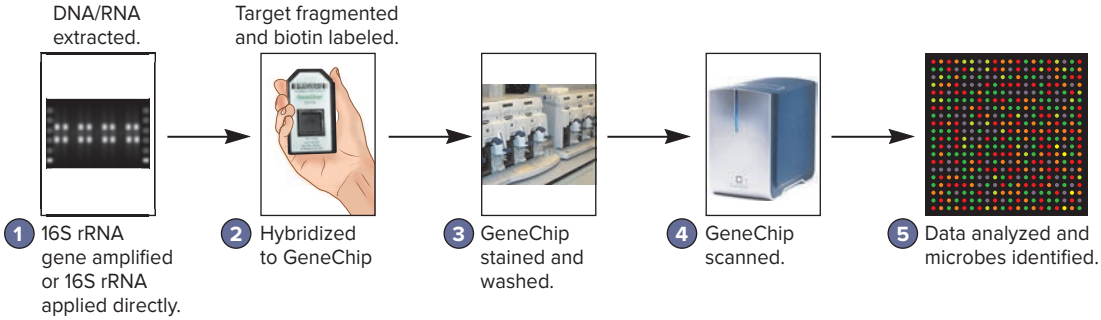


Figure 26.8 Phylochip Analysis. This form of microarray analysis enables the rapid evaluation of microbial diversity. (1) 16S rRNA genes or 16S rRNA molecules are obtained from the natural sample and labeled. (2) The 16S nucleic acids are hybridized to probes on a microarray called a phylochip. The probes consist of short regions of 16S rRNA genes obtained from preselected microbes. (3) The chip is washed to remove labeled 16S rRNA nucleic acids that could not hybridize because their complement is not found among the probes. (4) Laser beams scan the phylochip and (5) generate a series of spots of various colors and intensities that indicate the presence, absence, and, to a lesser extent, relative abundance of each microbe represented with an rRNA probe. These data are computer analyzed.

ribosomes (and therefore many rRNA targets), each cell has relatively few ribosomes and the amount of probe that binds is below detectable limits. A clever modification of the FISH technique amplifies the signal produced by each cell. The trick is to label the fluorescent probe with an enzyme that makes lots of fluorescent product when exposed to substrate. This is called *ca*talyzed *r*eported *d*eposition-FISH, or simply CARD-FISH. The enzyme attached to the oligonucleotide probe is usually horseradish peroxidase (HRP). After the oligonucleotides are hybridized to the sample, the HRP substrate tyramide is added. Tyramide is oxidized by HRP and the level of fluorescence is amplified. In this way, a single oligonucleotide probe bound to just one molecule of rRNA can generate an "enzyme boosted" fluorescent signal.

Specialized microarrays called *phylochips* (**figure 26.8**) and *geochips* can also be used to assess microbial diversity in natural samples without nucleotide sequencing. As described in section 18.5, microarrays are glass chips upon which single-stranded DNA probes are attached in a gridlike pattern. Each DNA probe is a single gene (or part of a gene), and its identity and location on the chip are recorded. Single-stranded nucleotides from an environmental sample are labeled with fluorescent tags and incubated with the chip so that complementary probe and sample sequences hybridize. Hybridization to each probe (gene) is measured by laser beams. Phylochip probes include 16S rRNA genes from thousands of bacteria and archaea, so ask the question "Who is in this environment?" Geochip probes represent over 24,000 genes involved in biogeochemical cycling (e.g., nitrogen fixation, phosphate uptake, etc.), thereby asking the question "What's happening in this environment?"

Metagenomic Analysis

As we learned in chapter 18, metagenomic analysis is the most efficient and comprehensive approach to surveying the genomes in a given environment (*see figure 18.8*). As such, metagenomics addresses both *macrodiversity*, or the number and types of individuals in an ecosystem, and *microdiversity*, defined as the genetic variation within a population. DNA from samples of soil, water, or other natural material (e.g., blood in a clinical setting) is extracted and used as template for shotgun metagenomics. Metagenomic analysis using high-throughput, next-generation sequencing is an important tool in studying microbial diversity. Computer software has been developed that analyzes shotgun metagenomic reads and quantifies the output such that the relative abundance of different microbial taxa can usually be assessed. However, because metagenomic sequences must be compared to reference genomes, identification is only possible for those microbes closely related to those with previously determined genome sequences. It should be stressed that novel microorganisms can (and have) been discovered using metagenomics. It requires the detection of enough overlapping sequence so that most (or ideally all) of the genome of the new microbe can be assembled. ◀ *DNA sequencing by synthesis (section 18.1); Metagenomics provides access to uncultured microbes (section 18.3)*

Metagenomic analysis generates a wealth of data. An investigator might be interested in a subset of genes (e.g., genes for nitrogen fixation), and the power of metagenomics is that it reveals the context in which these genes of interest exist. Generally speaking, the genetic diversity (microdiversity) of a community will include genes in high, medium, and low abundance. High frequency genes typically represent the core genome within the community. Genes found in moderate abundance are frequently involved in encoding secreted products for cell-cell interactions (e.g., siderophores, antibiotics, and quorum sensing molecules). Low frequency genes often encode surface structures that are species or strain specific (e.g., phage receptors, enzymes that assemble lipid A and O-antigen). These genes are often encoded on genomic islands or plasmids near transposases having been acquired by horizontal gene transfer, thereby making them part of a strain's pan-genome. Genotype diversity benefits occur at the population level, not the individual level for a number of reasons, such as increasing potential interactions and expanding the number of niches available for growth.

It is usually desirable to measure the number of template copies of a given gene (or genes) in a sample rather than simply

their relative number (e.g., high, medium, low). As discussed in chapter 17, real-time quantitative PCR (qPCR) provides a measure of template number and has been commonly used. More recently, **digital droplet PCR (ddPCR)** has become the method of choice when samples contain complex mixtures of molecules, including contaminating proteins that might otherwise inhibit the polymerase chain reaction. This is the case not only in environmental samples, but in many clinical samples. The DNA in each ddPCR sample is encapsulated into picoliter (10^{-12} liter) droplets containing either one or no template. At such a low dilution, the chance that any single droplet contains template follows a Poisson distribution, so the number of droplets containing template can be converted to a measurement of sample concentration. The ddPCR method enables detection and quantification of template at lower levels than is possible with qPCR. In addition, ddPCR provides an absolute measurement, not one based on a standard curve, as is the case with qPCR.

Comprehension Check

1. What macromolecules does DAPI stain? What are its advantages and disadvantages?
2. What is FISH? Describe an example in which the application of CARD-FISH might yield better results than traditional FISH.
3. Compare and contrast analysis of DNA extracted from a natural sample by shotgun metagenomics (described in chapter 18) and targeted PCR amplification of specific genes. What kinds of questions can be asked with each method?
4. How might phylochip analysis be useful in food safety investigations?

26.4 Assessing Microbial Community Activity

After reading this section, you should be able to:

- **a.** Describe why microelectrode measurements are important tools in the study of microbial community activity
- **b.** Compare the application of traditional stable isotope analysis with stable isotope probing
- **c.** Assess the advantages and disadvantages of measuring in situ mRNA abundance
- **d.** Differentiate metagenomics from metaproteomics
- **e.** Compare MAR-FISH with FISH

Many microbial ecologists want to understand how microbes drive and respond to the biogeochemical cycles essential for life on Earth. Thus they seek to identify and measure the flux of nutrients, the interactions between microbes that make these fluxes possible, and the influences of abiotic factors. Here we divide methods into biogeochemical and molecular techniques, but it should become clear that the results of one type of approach are often used to formulate hypotheses that will be tested by another.

▶| *Biogeochemical cycling sustains life on Earth (section 28.1)*

Biogeochemical Approaches

Perhaps the most fundamental microbial activity is growth. So the first question that may be asked is, "Are the microbes in this environment growing, and if so, at what rate?" Microbial growth rates in complex systems can be measured directly. Changes in microbial numbers are followed over time, and the frequency of dividing cells is also used to estimate production. Alternatively, the incorporation of radiolabeled thymidine (usually labeled with ^{3}H) into microbial biomass provides information about growth rates and microbial turnover. Because the nucleoside thymidine is found in DNA and not in RNA, its incorporation measures growth. This approach is most easily applied to aquatic microorganisms; it is more difficult to directly observe terrestrial microbial communities.

To determine community activity, it is necessary to have a good understanding of the microenvironment inhabited by the microbial community in question. In many ecosystems, these microhabitats are studied by **microelectrodes**—electrodes capable of measuring pH, oxygen tension, H_2, H_2S, or nitrogen species. The tips of these electrodes range from 2 to 5 µm wide and 100 to 200 µm long, enabling measurements to be taken as the tip penetrates the sample in increments of a millimeter or less. Because the concentration and distribution of electron donors (e.g., sulfide, ammonium) and acceptors (e.g., oxygen, nitrate, sulfate) are measured, microbial activities can be inferred. Microelectrodes have been widely used in the study of **microbial mats** (**figure 26.9**). These highly striated communities of photoautotrophs and chemotrophs develop in saline lakes, marine intertidal zones, and hot springs.

In determining community function, it is often instructive to explore the cycling of a particular elemental species. In aquatic and marine systems, the amount of carbon fixed by microbes is important. This is most commonly accomplished by measuring how much radioactive bicarbonate ($H^{14}CO_3$) is taken up by microbes. When bicarbonate dissolves in water, it dissociates and releases CO_2. Thus when added to a sample of microbes, its incorporation into particulate carbon (i.e., biomass) represents the *gross primary production;* that is, the total amount of carbon fixation without accounting for CO_2 lost through respiration. Specialized CO_2 traps can be added to the incubation vessel to measure its loss by respiration. When the amount of CO_2 respired is subtracted from the amount of CO_2 fixed into biomass, *net primary production* has been assessed.

In addition to radioactive isotope incorporation (such as that described for primary production), **stable isotope analysis** may be used. Isotopes are those forms of elements that differ in atomic weight because they bear different numbers of neutrons. Isotopes can be stable or unstable; only radioactive isotopes are unstable and decay. Stable isotopes are rare when compared to their lighter counterparts. For example, both ^{14}N and ^{15}N occur in nature and are stable, but ^{14}N is found in much greater abundance. Organisms discriminate between stable isotopes, and for any given element, the lighter (in this case, ^{14}N) is preferentially incorporated into biomass, a phenomenon known as **isotope fractionation.**